Computational Extension of OEIS Sequence A052075: A New Term and an Exhaustive Search to 4×10^{14}

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Abstract

We extend OEIS sequence A052075, which consists of primes whose successor prime appears as a contiguous decimal substring of the cube. We carry out an exhaustive computation over the interval from 6.9×10^{12} to 4×10^{14} using a segmented sieve of Eratosthenes, arbitrary-precision arithmetic, and an exact modular filter for trailing matches. The search examines approximately 1.2×10^{13} prime candidates in 37 contiguous segments and finds exactly one new term, $a(16) = 287\,902\,832\,031\,253$, the first addition since 2018. This computation yields the improved lower bound $a(17) > 4 \times 10^{14}$. We also prove a congruential restriction on trailing matches modulo 10 and discuss a heuristic model suggesting that the sequence is infinite.

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Key words and phrases: prime number, integer sequence, substring, cube of a prime, segmented sieve, computational number theory.

1 Introduction

A prime p belongs to sequence A052075 if the decimal representation of its successor prime $q = \text{nextprime}(p)$ appears as a contiguous substring of the decimal representation of p^3. De Geest [3] introduced the sequence in January 2000, and Handler [5] extended it to 15 terms in 2006. The most recent OEIS entry [6] records the lower bound $a(16) > 6.9 \times 10^{12}$ due to Resta [11], established in July 2018.

The computational difficulty has two distinct sources. First, the number of primes below X grows as $X/\ln X$ by the prime number theorem, so an exhaustive search to X requires testing on the order of $X/\ln X$ candidates. Second, computing p^3 for a 15-digit prime requires approximately 45-digit arithmetic, which overflows all standard fixed-width integer types. Specifically, for $p > 6.98 \times 10^{12}$ the cube exceeds the capacity of `__uint128_t`; the exact threshold is $(2^{128} - 1)^{1/3} \approx 6.981 \times 10^{12}$. Therefore arbitrary-precision arithmetic is required throughout the searched interval.

The present work resolves both difficulties. We use the GNU Multiple Precision Arithmetic Library (GMP) [4] for exact 45-digit cube computation, and we introduce a modular filter that detects trailing-position matches with very high efficiency. The resulting implementation exhausts $[6.9 \times 10^{12}, 4 \times 10^{14}]$ in about 66 elapsed hours on a commodity workstation.

The main results are as follows.

1. $a(16) = 287\,902\,832\,031\,253$ (Theorem 5), the first new term of A052075 since 2018.

2. $a(17) > 4 \times 10^{14}$ (Theorem 7), extending Resta's bound by a factor of approximately 58.

3. A congruential classification of trailing matches in Proposition 2.

4. A heuristic discussion suggesting that A052075 is infinite.

1.1 Notation

Throughout the paper, p denotes a prime, $q = \text{nextprime}(p)$ its successor prime, and $g = q - p$ the prime gap. We write $\lfloor x \rfloor$ for the floor function and $\pi(x)$ for the prime-counting function. All logarithms are natural unless otherwise stated. We use the notation $[L, R)$ for half-open integer intervals. For any positive integer n, we denote by $s(n)$ the decimal string of n without leading zeros, and by $d(n) = \lfloor \log_{10} n \rfloor + 1$ its digit count.

2 Algorithm

2.1 Segmented sieve

Candidate primes are enumerated by a segmented sieve of Eratosthenes [12]. The search interval $[L, R)$ is divided into macro-blocks of width $W = 1.28 \times 10^8$. Within each block, a standard byte-array sieve is applied after precomputing all primes up to $\sqrt{R}$ by a preparatory sieve. OpenMP [10] parallelism is applied across macro-blocks. With 16 logical threads on the test hardware, the implementation achieves a throughput of 45–59 $\times 10^6$ prime evaluations per second, or approximately 3.1×10^6 per thread, decreasing slowly with the digit count of p because of the growing cost of GMP arithmetic.

For primes $p \leq 6.98 \times 10^{12}$, the cube p^3 fits in `__uint128_t` and no GMP arithmetic is needed. For larger primes the implementation uses `mpz_t` arithmetic. Primality of candidates is guaranteed by the sieve construction; independent verification of the new term uses deterministic Miller–Rabin [1].

2.2 Substring matching

Given a prime p and $q = \text{nextprime}(p)$, a match occurs when the decimal string $s(q)$ is a contiguous substring of $s(p^3)$. Let $d_p = d(p)$ and $d_q = d(q)$ denote the digit counts of p and q, respectively. In the vast majority of cases in the searched interval, $d_q = d_p$. The exception occurs for the finitely many primes whose successor prime crosses a power-of-10 boundary (e.g., $\text{nextprime}(9\,999\,999\,999\,971) = 10\,000\,000\,000\,037$, so $d_q = d_p + 1$). The implementation uses the exact value of d_q in all cases. The digit count of p^3 satisfies $d(p^3) \in \{3d_p - 2,\ 3d_p - 1,\ 3d_p\}$; we write $D = d(p^3)$ for the exact value.

We call the match *trailing* if $s(q)$ occupies the last d_q digits of $s(p^3)$, that is, if $p^3 \equiv q \pmod{10^{d_q}}$. We call it *leading* if $s(q)$ occupies the first d_q digits, and *interior* otherwise. Among the 16 known terms, trailing matches occur for $n = 10, 12, 15, 16$, and leading matches occur for $n = 1, 2, 13$; see Table 1.

2.3 Modular filter for trailing matches

The trailing condition $p^3 \equiv q \pmod{10^{d_q}}$ can be rewritten purely in terms of p and the prime gap $g = q - p$ as

$$p^3 \equiv p + g \pmod{10^{d_q}}. \tag{1}$$

This simplifies to

$$p(p^2 - 1) \equiv g \pmod{10^{d_q}}. \tag{2}$$

Lemma 1. *A prime p with* nextprime$(p) = p + g$ *satisfies the trailing-match condition if and only if $p(p^2-1) \equiv g \pmod{10^{d_q}}$, where $d_q = \lfloor \log_{10}(p+g) \rfloor + 1$.*

Proof. The trailing condition $p^3 \equiv q \pmod{10^{d_q}}$ is equivalent to $p^3 - q \equiv 0 \pmod{10^{d_q}}$. Substituting $q = p + g$ gives $p^3 - p - g \equiv 0 \pmod{10^{d_q}}$, which is equivalent to $p(p^2 - 1) \equiv g \pmod{10^{d_q}}$. □

The filter of Lemma 1 yields an immediate structural consequence.

Proposition 2. *Let $p > 5$ be a prime with* nextprime$(p) = p + g$*. If p admits a trailing match, then*

$$g \equiv \begin{cases} 0 \pmod{10}, & \text{if } p \equiv 1 \pmod{10}, \\ 4 \pmod{10}, & \text{if } p \equiv 3 \pmod{10}, \\ 6 \pmod{10}, & \text{if } p \equiv 7 \pmod{10}, \\ 0 \pmod{10}, & \text{if } p \equiv 9 \pmod{10}. \end{cases}$$

Proof. By Lemma 1, a trailing match requires $p(p^2-1) \equiv g \pmod{10^{d_q}}$, and hence $p(p^2 - 1) \equiv g \pmod{10}$. Since $p > 5$ is prime, $p \bmod 10 \in \{1, 3, 7, 9\}$. We compute $p(p^2 - 1) \bmod 10$ for each residue class:

$$\begin{aligned} p \equiv 1 \pmod{10} &\implies p(p^2-1) \equiv 1 \cdot 0 = 0 \pmod{10}, \\ p \equiv 3 \pmod{10} &\implies p(p^2-1) \equiv 3 \cdot 8 = 24 \equiv 4 \pmod{10}, \\ p \equiv 7 \pmod{10} &\implies p(p^2-1) \equiv 7 \cdot 48 = 336 \equiv 6 \pmod{10}, \\ p \equiv 9 \pmod{10} &\implies p(p^2-1) \equiv 9 \cdot 80 = 720 \equiv 0 \pmod{10}. \end{aligned}$$

This proves the claim. □

Remark 3. Proposition 2 is consistent with all four trailing matches among the 16 known terms. For $a(10)$ we have $p \equiv 7$ and $g = 56 \equiv 6 \pmod{10}$; for $a(12)$ we have $p \equiv 7$ and $g = 96 \equiv 6 \pmod{10}$; for $a(15)$ we have $p \equiv 9$ and $g = 40 \equiv 0 \pmod{10}$; and for $a(16)$ we have $p \equiv 3$ and $g = 24 \equiv 4 \pmod{10}$.

Remark 4. The classification can be extended to higher moduli by applying the Chinese remainder theorem to the factors 2^k and 5^k of 10^k. For example, modulo 100 one obtains a finer partition of admissible residue pairs ($p \bmod 100, g \bmod 100$), further restricting the candidates that can produce a trailing match.

The modular filter of Lemma 1 is exact for trailing matches: it accepts a candidate if and only if a trailing match occurs. In the implementation, the filter is evaluated as a fast 128-bit modular computation for every prime p. Candidates that pass the filter are flagged as potential trailing matches and recorded immediately, without waiting for the full substring scan. Candidates that do not pass the filter are not discarded: every prime in the search interval proceeds to the full computation of p^3 and the complete scan of $s(p^3)$ for $s(q)$, so leading and interior matches are not missed.

As a rough heuristic sanity check, for 15-digit primes the average prime gap near 10^{15} is approximately $\ln(10^{15}) \approx 35$ by the prime number theorem. This gives a theoretical expected number of trailing-match candidates of order $(35/10^{15}) \times 1.2 \times 10^{13} \approx 0.4$ over the searched interval, consistent with finding exactly one such candidate ($a(16)$) and zero spurious trailing hits.

3 Computational results

3.1 Oracle verification

All 15 previously known terms were verified independently at the start of every search segment. For each term $a(n)$, we confirmed the primality of p, the correctness of nextprime(p), the exact value of p^3, and the substring match at the stated position. Table 1 lists the complete oracle set, including the new term $a(16)$.

n	$a(n) = p$	nextprime(p)	pos	len	type
1	11	13	0	4	L
2	101	103	0	7	L
3	2 239	2 243	2	11	I
4	34 297	34 301	2	14	I
5	43 789	43 793	4	14	I
6	53 549	53 551	1	15	I
7	535 487	535 489	1	18	I
8	59 897 017	59 897 039	14	24	I
9	430 784 719	430 784 737	3	26	I
10	2 549 592 677	2 549 592 733	19	29	T
11	2 837 138 669	2 837 138 677	1	29	I
12	97 969 345 967	97 969 346 063	22	33	T
13	100 000 000 019	100 000 000 057	0	34	L
14	328 096 840 219	328 096 840 223	20	35	I
15	4 110 739 763 869	4 110 739 763 909	25	38	T
16	287 902 832 031 253	287 902 832 031 277	29	44	T

Table 1: Complete known terms of A052075. Column pos gives the 0-based starting position of nextprime(p) in $s(p^3)$; column len gives the digit count $D = d(p^3)$; column type classifies the match as leading (L), trailing (T), or interior (I). Full values of p^3 for all 16 terms are available in the oracle CSV file at the GitHub repository.

3.2 New term: $a(16)$

The exhaustive search of the interval $[6.9 \times 10^{12}, 4 \times 10^{14}]$ yields exactly one new term.

Theorem 5. *We have* $a(16) = 287\,902\,832\,031\,253$.

Proof. We verify the four defining properties.

1. $p = 287\,902\,832\,031\,253$ is prime, as confirmed by `sympy.isprime`.

2. $q = \text{nextprime}(p) = 287\,902\,832\,031\,277$, with gap $g = 24$, as confirmed by `sympy.nextprime`. This also shows that every integer in the open interval (p, q) is composite.

3. The cube is the 44-digit integer

$$p^3 = 23\,863\,701\,656\,637\,949\,988\,435\,953\,863\,287\,902\,832\,031\,277,$$

as confirmed by GMP and by Python `int` arithmetic.

4. The decimal string of q occupies positions 29 through 43 of $s(p^3)$, using 0-based indexing. Thus the last 15 digits of the cube equal q, so this is a trailing match.

The campaign summary in Tables 2 and 3 documents the exhaustiveness of the search on $[6.9 \times 10^{12}, 4 \times 10^{14}]$. □

Remark 6. The modular filter in (2) confirms $p(p^2 - 1) \equiv 24 \pmod{10^{15}}$. Indeed, $p \equiv 287\,902\,832\,031\,253 \pmod{10^{15}}$, so $p(p^2 - 1) \bmod 10^{15} = 24$.

3.3 Improved lower bound

Theorem 7. *We have $a(17) > 4 \times 10^{14}$.*

Proof. This is a computational theorem: the conclusion rests on the verified exhaustive search documented in Tables 2 and 3. That search covers every prime in $[6.9 \times 10^{12}, 4 \times 10^{14}]$ and detects no match of any type—leading, trailing, or interior—other than the single term $a(16)$ found in segment S28. By Theorem 5, this term equals $287\,902\,832\,031\,253$, which lies in $[6.9 \times 10^{12}, 4 \times 10^{14}]$. Therefore $a(17)$, the next term after $a(16)$, must exceed 4×10^{14}. In particular, segments S29 through S37 cover $[2.92 \times 10^{14}, 4 \times 10^{14}]$ with zero hits, confirming the absence of any additional term in that subinterval. □

3.4 Campaign summary

The search was executed as 37 contiguous segments on a workstation with an AMD Ryzen 9 7940HS processor (8 cores, 16 threads), 64 GB DDR5 RAM, running Linux Mint 22.3. Tables 2 and 3 summarize the key parameters of each segment.

The total elapsed wall-clock time is 237 868 seconds, or about 66.1 hours. With 16 OpenMP threads, the average throughput is approximately 50×10^6 prime evaluations per second, corresponding to roughly 3.1×10^6 evaluations per thread. The calendar elapsed time is approximately 7.2 days, from 2026-03-24 to 2026-03-31, reflecting discontinuous use of the workstation.

Seg	Range $[L, R) \times 10^{12}$	Primes ($\times 10^9$)	Elapsed (s)	Hits	Date
S01	[6.9, 14.9)	266.7	4522	0	2026-03-24
S02	[14.9, 22.9)	261.8	4576	0	2026-03-24
S03	[22.9, 30.9)	258.7	4624	0	2026-03-24
S04	[30.9, 38.9)	256.6	4688	0	2026-03-25
S05	[38.9, 46.9)	254.9	4687	0	2026-03-25
S06	[46.9, 54.9)	253.5	4647	0	2026-03-25
S07	[54.9, 62.9)	252.3	4695	0	2026-03-25
S08	[62.9, 70.9)	251.3	4633	0	2026-03-25
S09	[70.9, 78.9)	250.4	4648	0	2026-03-25
S10	[78.9, 86.9)	249.6	4714	0	2026-03-25
S11	[86.9, 94.9)	248.9	4737	0	2026-03-26
S12	[94.9, 100.0)	158.3	3051	0	2026-03-26
S13	[100.0, 112.0)	371.6	7030	0	2026-03-26
S14	[112.0, 124.0)	370.4	7198	0	2026-03-26
S15	[124.0, 136.0)	369.3	7074	0	2026-03-26
S16	[136.0, 148.0)	368.3	7172	0	2026-03-27
S17	[148.0, 160.0)	367.3	7152	0	2026-03-27
S18	[160.0, 172.0)	366.5	7432	0	2026-03-27
S19	[172.0, 184.0)	365.7	7141	0	2026-03-27

Table 2: Campaign summary, segments S01–S19. Segment widths are 8×10^{12} for S01–S11 and 12×10^{12} for S13–S19 (S12 has width 5.1×10^{12}). The time column reports elapsed wall-clock time of the C process with 16 OpenMP threads. Prime counts are rounded to one decimal place; the total in Table 3 is computed from unrounded raw counts.

3.5 Independent verification

The new term $a(16) = 287\,902\,832\,031\,253$ was verified independently in Python 3 with `sympy` (version 1.14.0) as follows.

1. `sympy.isprime(287902832031253)` returns `True`.
2. `sympy.nextprime(287902832031253)` returns `287902832031277`.
3. `287902832031253**3` evaluates to `23863701656637949988435953863287902832031277`.

Seg	Range $[L, R) \times 10^{12}$	Primes ($\times 10^9$)	Elapsed (s)	Hits	Date
S20	[184.0, 196.0)	365.0	7302	0	2026-03-28
S21	[196.0, 208.0)	364.3	7144	0	2026-03-28
S22	[208.0, 220.0)	363.7	7290	0	2026-03-28
S23	[220.0, 232.0)	363.1	7343	0	2026-03-28
S24	[232.0, 244.0)	362.5	7471	0	2026-03-29
S25	[244.0, 256.0)	362.0	7377	0	2026-03-29
S26	[256.0, 268.0)	361.5	7387	0	2026-03-29
S27	[268.0, 280.0)	361.0	7406	0	2026-03-29
S28	[280.0, 292.0)	360.5	7480	1	2026-03-30
S29	[292.0, 304.0)	360.1	7412	0	2026-03-30
S30	[304.0, 316.0)	359.6	7970	0	2026-03-30
S31	[316.0, 328.0)	359.2	7352	0	2026-03-30
S32	[328.0, 340.0)	358.8	7345	0	2026-03-30
S33	[340.0, 352.0)	358.5	7434	0	2026-03-31
S34	[352.0, 364.0)	358.1	7493	0	2026-03-31
S35	[364.0, 376.0)	357.7	7361	0	2026-03-31
S36	[376.0, 388.0)	357.4	7434	0	2026-03-31
S37	[388.0, 400.0)	357.1	7446	0	2026-03-31
Total	[6.9, 400.0)	12 031.9	237 868	1	

Table 3: Campaign summary, segments S20–S37, and totals. Elapsed times in seconds; the total row gives the sum over all 37 segments. Prime counts are rounded to one decimal place; the total is computed from unrounded raw counts.

4. `str(287902832031277) in str(287902832031253**3)` returns `True`, at position 29 of the 44-digit string.

All four conditions are satisfied. For numbers below $2^{64} \approx 1.84 \times 10^{19}$, the `sympy.isprime` function performs a deterministic primality test based on trial division and deterministic Miller–Rabin with verified witness sets [13], so the verification in step 1 is conclusive rather than probabilistic. The value $p \approx 2.88 \times 10^{14}$ lies well below this bound.

4 Probabilistic heuristic

4.1 Expected density of terms

For a prime p with $d = d(p)$ decimal digits, let $q = \text{nextprime}(p)$, which also has d digits for all sufficiently large p in a fixed decade. The cube p^3 has $D \in \{3d-2,\ 3d-1,\ 3d\}$ digits. For a rough estimate we take $D \approx 3d$.

The probability that a uniformly chosen d-digit string $s(q)$ appears as a substring of a uniformly chosen D-digit string $s(p^3)$ is approximately

$$P_{\text{sub}} \approx \frac{D-d+1}{10^d} \approx \frac{2d}{10^d}, \tag{3}$$

since there are approximately $D - d + 1 \approx 2d$ possible starting positions and each position matches with probability 10^{-d}.

The number of d-digit primes is approximately $\pi(10^d) - \pi(10^{d-1}) \approx 8 \cdot 10^{d-1}/d$ (a standard rough estimate from the prime number theorem). The expected number of terms of A052075 in the d-digit decade is therefore

$$E_d \approx \frac{8 \cdot 10^{d-1}}{d} \cdot \frac{2d}{10^d} = 1.6. \tag{4}$$

The strings $s(q)$ and $s(p^3)$ are not independent, because both are determined by p. The study of digit strings in prime-related sequences [2] suggests that such arithmetic correlations can produce a moderate downward bias. Empirically, the 15 known terms in the 12 decades from $d = 2$ through $d = 13$ give an observed density of $15/12 = 1.25$ terms per decade. The ratio of this observed density to the crude estimate $E_d \approx 1.6$ yields a correction factor of

$$\frac{15}{12 \times 1.6} \approx 0.78.$$

Applying this correction factor to the crude estimate gives an adjusted expected count of $0.78 \times 1.6 = 1.25$ terms per decade, consistent with the observed density.

4.2 Implication for infinitude

Since the adjusted estimate of approximately 1.25 terms per decade remains bounded away from zero across decades, the heuristic model suggests that sufficiently large decades should continue to contain terms of A052075 with positive probability. Under an additional assumption that occurrences in well-separated decades are approximately independent—which is plausible but unproved—this points toward infinitely many terms. This is only a heuristic argument. The events are not independent across primes, and the approximation (3) treats $s(q)$ and $s(p^3)$ as independent, which is false. The discussion serves only as motivation for the following conjecture.

Conjecture 8. *The sequence* A052075 *is infinite.*

A rigorous proof of Conjecture 8 seems to be beyond current methods, since the problem mixes the distribution of prime gaps with the non-algebraic structure of decimal representations.

5 Conclusion and open problems

We determine $a(16) = 287\,902\,832\,031\,253$, the first new term of A052075 since 2018, and establish the improved lower bound $a(17) > 4 \times 10^{14}$. The computation requires about 66 elapsed hours on a commodity workstation and is made feasible by GMP arithmetic, an exact modular filter for trailing-position candidates, and an OpenMP-parallelized segmented sieve.

The main open problems are as follows.

Open Problem 9. *Determine* $a(17)$. *Under the heuristic model, the interval* $[4 \times 10^{14}, 10^{15}]$ *has logarithmic length* $\log_{10}(2.5) \approx 0.398$ *decades. Applying the observed density of 1.25 terms per decade gives an expected count of about* $0.398 \times 1.25 \approx 0.50$*, so the Poisson-model probability of at least one term in this interval is approximately* $1 - e^{-0.50} \approx 0.39$.

Open Problem 10. *Prove or disprove Conjecture 8.*

Open Problem 11. *Characterize which positions, namely leading, trailing, and interior, can support a match, and determine whether trailing matches are denser than interior matches. Among the 16 known terms, 4 are trailing, 3 are leading, and 9 are interior.*

Open Problem 12. *Investigate the analogous sequences for* p^k *with* $k \geq 2$*, including the square analogue* A052073 *[7], the related sequence* A321796 *[8], in which* $\mathrm{prevprime}(p)$ *is a substring of* p^3*, and the sequence* A381969 *[9], in which* $\mathrm{prevprime}(p)$ *is a substring of* p^2.

6 Acknowledgments

The author thanks the OEIS contributors whose earlier computations established the initial terms and the previous lower bound for A052075.

7 Data availability

The complete dataset, including the oracle CSV file, per-segment logs, campaign summary, and machine-readable results, is available at the GitHub repository `https://github.com/carcorti/A052075` and at the corresponding Zenodo archive `https://doi.org/10.5281/zenodo.19421914`. The results are also available through the OEIS entry for A052075.

8 Code availability

The C source file `a052075_v4.c` (version 4 of the search program), the associated Makefile, and a Python verification script `verify_a052075.py` are available at the GitHub and Zenodo repositories listed in the previous section.

9 AI use disclosure

The author used large language model tools during the preparation of this work. The AI tools assisted with code review, verification of computational results, and identifying weaknesses in the exposition. No prose in this manuscript was generated by a large language model. All mathematical arguments, computational claims, and final wording of the manuscript were produced and verified by the author, who takes full scientific responsibility for the paper.

10 Sequences

The sequences appearing in this paper are as follows.

- A052073, the square analogue of A052075 mentioned in Open Problem 12.

- A052075, primes p such that the successor prime nextprime(p) appears as a contiguous decimal substring of p^3.
- A321796, primes p such that the predecessor prime prevprime(p) appears as a contiguous decimal substring of p^3, mentioned in Open Problem 12.
- A381969, primes p such that the predecessor prime prevprime(p) appears as a contiguous decimal substring of p^2, mentioned in Open Problem 12.

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